Are Hospitals Where the People Are? A National Inhomogeneous K-Function Test of U.S. Hospital Distribution Against Ambient Population

Matthew R. Lehnert, Axle Informatics (Matthew.Lehnert@axleinfo.com)

July 2026

Abstract

Background. Whether hospitals are located where people need them is usually studied with catchment-based accessibility scores and residential census population. We reframe the question as a formal spatial point-process test and, critically, ask it against *ambient* population, where people are over a 24-hour day, rather than only where they sleep.

Methods. We characterize hospital supply in every US commuting zone (county clusters that tile the entire country, rural included) on three orthogonal axes: **concentration** (is supply clustered beyond population?), measured by an inhomogeneous Ripley's K-function with a population-proportional null for both hospital locations and bed capacity; **coverage** (are people far from care?), measured by population-weighted distance to the nearest hospital against the same null, crediting hospitals across zone boundaries; and **sufficiency** (is there enough capacity?), measured by beds per 1,000. Concentration and coverage are tested against Monte-Carlo simulations in which population is fixed and hospitals are resampled, with a Myllymaki global envelope test controlling error within and across zones. Hospitals come from HIFLD (n = 7,966). Ambient population is ORNL LandScan; residential population is NASA SEDAC GPWv4.11.

Results. The demand surface reverses the conclusion. Against residential population, 15 of 262 hospital-dense zones are significantly over-concentrated (family-wise p = 0.002) and none are under-served; against ambient population, that over-concentration disappears entirely (0 zones, p = 0.47) and instead 6 zones are significantly under-served (p = 0.001). Over-concentration and under-service do not co-occur in the same zone (a pattern that is partly structural, since concentration is tested only in hospital-dense zones), and 44 of 598 zones change diagnosis between the two surfaces. A WorldPop control (modeled but residential) shows the reversal tracks day-versus-night timing, not raster construction, and the findings are stable to the hospital-count threshold, the coverage buffer, a re-run on 939 CBSA windows, and neutralizing the population LandScan places at the hospital sites themselves.

Conclusions. Whether US hospital supply looks over-provided or underserving depends on whether demand is counted where people sleep or where they are by day. Accessibility analyses that default to residential population may systematically mistake the nature of local hospital-access inequity.

Keywords: inhomogeneous K-function; spatial point patterns; healthcare access; hospital capacity; ambient population; LandScan; hospital deserts.

Introduction

The problem

Access to hospital care is unevenly distributed across the United States, and the consequences of that unevenness, including delayed treatment, worse outcomes, and rural hospital closures, are well documented (Branas et al., 2005; Carr et al., 2017; Hsia & Shen, 2011). A foundational question underlies any equity claim: are hospitals located in proportion to the population that needs them? Where they are not, two distinct failures appear. The first is over-concentration, where facilities accumulate beyond what local population warrants, producing urban redundancy and competitive clustering. The second is under-service, where population is present but nearby hospital provision is thin, producing access deserts.

The dominant paradigm and its limits

Most quantitative work on hospital access uses catchment-based accessibility scores, principally the two-step floating catchment area (2SFCA) family and gravity models, which produce a per-location accessibility index from supply, demand, and travel impedance. These methods are effective for ranking places, but they have three properties that motivate an alternative. They require modeling travel impedance and a distance-decay function whose form is a researcher choice. They yield a descriptive score rather than a formal hypothesis test of proportionality. And they almost universally use residential (nighttime) census population as the demand surface.

Our reframing: a point-process test of proportionality

We treat hospital locations as a spatial point pattern and ask directly whether that pattern is consistent with a process that places hospitals in proportion to population. The natural instrument is the inhomogeneous Ripley's K-function, the second-order summary of a point pattern relative to a specified first-order intensity. By setting that intensity proportional to population, the null becomes a statement that hospitals are distributed as an inhomogeneous Poisson process driven by population, and departures from the Monte-Carlo envelope are interpretable as population-relative over- or under-clustering. A homogeneous (complete spatial randomness) test would only rediscover that hospitals sit where people are; the inhomogeneous formulation is what makes the proportionality question answerable.

A methodological point distinguishes this design from kernel-based inhomogeneous analyses. Our intensity is supplied externally by a population raster rather than estimated from the hospital points themselves. This removes the bandwidth selection, intensity-floor, and isolated-point instability that affect point-estimated inhomogeneous intensities, and it makes the null an explicit, auditable demand surface rather than a smoothing artifact.

Why ambient population

Conventional access analyses use where people sleep. People have medical emergencies, however, where they work, shop, and commute. Ambient (24-hour average) population, operationalized here with ORNL LandScan, is a more appropriate demand surface for acute hospital care. We make the ambient versus residential choice a first-class object of study, running the identical test under both demand surfaces and reporting where the equity verdict changes. This choice turns out not to refine the picture but to reverse it: measured against residential population, US hospitals appear significantly over-concentrated, but that over-concentration vanishes against ambient population, which instead reveals access gaps the residential surface hides.

Three axes, not one number

Concentration alone does not capture access. A region can have hospitals arranged proportional to population yet leave a remote population stranded, or have adequate buildings but capacity concentrated in one center, or simply have too few beds for its people. We therefore measure three orthogonal axes: **concentration** (is supply clumped beyond population?), **coverage** (are people far from care?), and **sufficiency** (is there enough capacity per person?). Each answers a distinct question that a single accessibility score conflates, and their combination yields a per-zone diagnosis that maps to a distinct policy response.

Contributions

1. A formal, point-process test of whether US hospital **locations** and **bed capacity** follow population, applied nationally across all US commuting zones, which tile the entire country (rural included) and are exogenous to hospital locations, with family-wise error control within and across zones.
2. The first use, to our knowledge, of **ambient** population as the point-process null for hospital distribution, with a direct ambient versus residential contrast that we report at the level of changed verdicts.
3. A **coverage** test of population-weighted distance to care that credits cross-border hospitals, and a **sufficiency** measure of beds per capita, combined with concentration into a **diagnostic typology** (shortage, maldistribution, geographic gap, redundancy) that distinguishes "too few" from "in the wrong place" from "people stranded."
4. A reproducible, validated (Type-I calibrated), open implementation with a Nextflow pipeline.

Roadmap

Section 2 reviews related work and states the gap. Section 3 describes the hospital, population (ambient and residential), and commuting-zone data. Section 4 develops the three axes, the population-proportional null, the cross-border coverage construction, and the global envelope tests. Section 5 reports results. Section 6 concludes with limitations and policy implications.

Related Work

This study asks whether US hospitals are sited in proportion to the population they serve, and answers that question with an inhomogeneous, univariate Ripley's K-function in which the null intensity is fixed proportional to an externally supplied *ambient* population surface and tested against a global, family-wise-corrected Monte Carlo envelope. The work sits at the intersection of five literatures: spatial point-pattern methodology, mark-weighted concentration measures from economic geography, the dominant accessibility-modeling paradigm in health geography, the population-data literature on ambient versus residential counts, and global-envelope inference. We review each in turn, then state what is established and where our contribution lies.

Spatial point-pattern methods for facility location

Ripley's K-function and its variants are the standard second-order summary statistics for testing whether a set of points is clustered, regular, or consistent with complete spatial randomness (CSR). The methodological step that matters for our purposes is the *inhomogeneous* K-function, $K_{\text{inhom}}(r)$, introduced by A. J. Baddeley et al. (2000), which generalizes Ripley's statistic to processes with non-constant intensity $\lambda(u)$ by weighting each point pair by $1/[\lambda(x_i)\lambda(x_j)]$ under second-order intensity-reweighted stationarity. This estimator, together with Monte Carlo envelope testing, is implemented in the `spatstat` ecosystem (A. Baddeley et al., 2015) and is the machinery our study uses. Supplying $\lambda(u) \propto$ population converts the abstract question of whether points are clustered into the substantive question of whether they are clustered relative to where people are.

The justification for treating population as the baseline intensity comes from Diggle's case-control tradition. Diggle et al. (2007) extended second-order analysis to non-stationary patterns by contrasting disease case locations against control points sampled from the population at risk, so that residual clustering reflects elevated risk rather than the geography of where people live. We adopt the same inferential logic, with population as the denominator against which point intensity is judged. Diggle and colleagues realize the population through a sampled control point pattern, whereas we supply an explicit gridded population raster as $\lambda(u)$ and apply the framework to facilities rather than disease cases.

Applied point-pattern studies of healthcare facilities exist, but almost all use the homogeneous CSR null rather than a population-weighted one. Recent work analyzing hospitals and elder-care institutions in Wuhan applied the standard Ripley's K with Monte Carlo envelopes to argue for spatial inequity (Li et al., 2025), and network-constrained auto- and cross-K analyses of hospitals and pharmacies in Nanjing similarly test against CSR (Ni et al., 2016). These studies share our domain of hospital points, envelope testing, and an equity question, but not our null. Testing against CSR detects only whether facilities cluster in space, not whether they cluster out of proportion to the population they ostensibly serve, which is the question our analysis addresses.

Capacity- and mark-weighted concentration measures

Treating each hospital as a dimensionless point discards the fact that a 600-bed academic medical center and a 15-bed critical-access hospital are not equivalent units of supply. The economic-geography literature has confronted this same problem for firms. The theoretical basis is the mark-weighted K-function of Penttinen et al. (1992), in which point pairs contribute weights that depend on their marks; under random labelling the weighted statistic reduces to the unweighted K, so departures isolate mark-dependent structure. Building on this, Giuliani et al. (2014) weight Ripley's K by firm dimension (employees, output, capital) and show that size-weighting can materially change whether an industry is judged to be concentrated. This is the closest published analog to our bed-weighted variant; substituting hospital beds for firm employees yields a capacity-weighted hospital K. A parallel distance-based tradition develops continuous-space, baseline-relative concentration measures that mirror our approach of a reference population as the benchmark with weights for unit size: Besag's L and Diggle-Chetwynd's D imported into economic geography by Marcon & Puech (2003), the relative and weight-flexible M-function of Marcon & Puech (2010), and the kernel-based localization test with Monte Carlo confidence bands of Duranton & Overman (2005). These methods are established. We did not find their transfer to capacity-weighted hospital point patterns benchmarked against population.

Accessibility paradigms: 2SFCA, E2SFCA, and gravity models

The dominant paradigm for studying spatial equity of healthcare in the US is not point-pattern analysis but accessibility modeling. The two-step floating catchment area (2SFCA) method of Luo & Wang (2003) computes provider-to-population ratios within travel-time catchments and sums them to residential locations; the enhanced E2SFCA of Luo & Qi (2009) adds distance-decay weights within nested travel-time zones; and gravity-model formulations generalize the distance-decay structure. Wang (2012) provides the canonical methodological review, and the systematic review of Stacherl & Sauzet (2023) confirms that the 2SFCA family dominates both methodological development and empirical application. These methods produce a per-location accessibility score and are the natural competitor to our framing. Two features distinguish our work. First, accessibility models answer how much supply each demand location can reach, whereas the inhomogeneous K answers a complementary, scale-explicit question of the scales across which supply is over- or under-concentrated relative to demand, with a formal hypothesis test rather than a descriptive score. Second, the 2SFCA literature almost universally uses residential population as the demand denominator. Substituting an ambient denominator is rare, a point we return to below.

Ambient versus residential population

LandScan, introduced by Dobson et al. (2000), was conceived as an ambient population product, a 24-hour average of human presence rather than a nighttime residential count, for estimating populations at risk. LandScan USA later distinguished nighttime-residential from daytime-ambient distributions within a single product family (Bhaduri et al., 2007), and the recent LandScan HD methodology defines ambient population as average human presence between day and night and names spatial accessibility to vital resources among its applications (Sims et al., 2025). The complementary residential surface we contrast against, NASA SEDAC / CIESIN's Gridded Population of the World version 4, is documented by Doxsey-Whitfield et al. (2015). The argument that the choice of denominator (ambient versus residential) changes equity conclusions is well established in exposure science and crime analysis, but it has barely entered healthcare-access work. The nearest health-access precedent is Gligorić et al. (2023), who contrast mobility-revealed accessibility (where people actually travel) against residence-based potential accessibility and find the two diverge by more than an hour of travel time in dozens of countries, evidence that residence-based denominators systematically misstate real access. That study uses mobility traces, not an ambient population grid, and operates within the travel-time accessibility paradigm rather than a point-process test. To our knowledge no prior work substitutes an ambient population raster as the null intensity of a hospital point-pattern test.

Global envelopes and multiple-testing rigor

A recurring weakness of envelope-based point-pattern inference is that pointwise Monte Carlo envelopes control the Type I error only at a single distance r, even though the test curve is inspected across an entire range of r, leaving an uncontrolled multiple-comparison problem. Myllymäki et al. (2017) addressed this with global (rank-based) envelope tests that fix a global α a priori, return an exact p-value, and remain graphically interpretable; Mrkvička et al. (2017) extended the framework to control the family-wise error across several function-valued tests, and the `GET` package (Myllymäki & Mrkvička, 2024) implements these tests, including the extreme rank length (ERL) ordering we use. Because our design fits one K_{inhom} test per analysis window across the entire country, family-wise correction is not optional, and the ERL global envelope is the appropriate instrument. The applied US hospital point-pattern studies cited above rely on uncorrected pointwise envelopes.

US hospital and trauma access: the substantive backdrop

The substantive motivation, urban over-concentration coexisting with underserved gaps, is documented most influentially by Branas et al. (2005), who found that 84.1% of US residents reached a Level I/II trauma center within 60 minutes, yet tens of millions of mostly rural residents had no such access while many urban residents could reach 20 or more centers within the hour. Carr et al. (2017) confirmed and refined this with a block-group analysis tying access to income and urbanicity, and Hsia & Shen (2011) showed access erosion concentrated among vulnerable populations. Optimization studies such as the anti-covering models of Chea et al. (2022) explicitly target overlapping urban service areas to redeploy capacity to underserved areas. These works establish that the over- and under-concentration phenomenon is real and policy-relevant, but they quantify it through travel-time coverage or facility-siting optimization, not through a formal point-process test of proportionality to population. We adopt USDA Commuting Zones (Fowler et al., 2016; Tolbert & Sizer, 1996) as analysis windows because they tile the entire country (including rural areas the metropolitan and micropolitan scheme omits), are defined by commuting flows rather than administrative or hospital-related boundaries, and are therefore exogenous to the facility locations under test, a property that matters when the window itself could otherwise bias a spatial test. Hospital points, bed counts, and trauma designations are drawn from the HIFLD national dataset (U.S. Department of Homeland Security, 2024).

Gap and contribution

We are clear about what is not novel. The inhomogeneous K-function (A. J. Baddeley et al., 2000), the use of population as a point-process baseline (Diggle et al., 2007), mark and capacity weighting of K (Giuliani et al., 2014; Penttinen et al., 1992), global family-wise-corrected envelope tests (Myllymäki et al., 2017; Myllymäki & Mrkvička, 2024), LandScan ambient population (Dobson et al., 2000), and the documented reality of US hospital and trauma over- and under-concentration (Branas et al., 2005) are all established. We claim none of these as our own.

Our contribution is their specific synthesis and application, which we did not find assembled anywhere in the literature. First, we build an inhomogeneous, capacity-weightable Ripley's K of hospital point locations whose null intensity is set proportional to an ambient (LandScan, 24-hour) population surface, rather than the residential denominator that the 2SFCA accessibility paradigm uses by default (Luo & Wang, 2003; Stacherl & Sauzet, 2023) and rather than a kernel-estimated, data-driven intensity, and we contrast it directly against a residential surface (GPW v4) (Doxsey-Whitfield et al., 2015). Second, we run a national application across all US Commuting Zones, with a family-wise-corrected global (ERL) envelope test, so that the resulting map of over- and under-served zones carries a multiplicity-controlled significance statement. Third, we add local decomposition (`localKinhom`) to localize deviations and a bed-weighted capacity variant. The applied hospital point-pattern studies use a CSR null and uncorrected envelopes; the accessibility literature uses residential denominators and produces scores rather than tested proportionality; the ambient-versus-residential contrast has been made in exposure and revealed-mobility settings (Gligorić et al., 2023) but not as the null of a hospital point-process test. The gap we fill is therefore not methodological invention but the principled combination of an established estimator, an ambient null, capacity weighting, and rigorous global inference, applied at national scale to whether US hospital supply is spatially proportional to the population it serves.

Data

All three inputs are public. Acquisition is scripted in `code/acquire/`; the exact provenance, vintages, and filtering are recorded below and in each derived file's metadata.

Hospital locations (HIFLD)

Hospital point locations come from the Homeland Infrastructure Foundation-Level Data (HIFLD) "Hospitals" layer (DHS / Oak Ridge National Laboratory). The HIFLD Open portal was deactivated on 2025-08-26, so we source the layer from the DataLumos archive (ICPSR project 239108).

- **Count.** 8,340 facilities, of which 7,966 are `STATUS == OPEN`; the 374 closed facilities are excluded.
- **Geometry.** Point locations in EPSG:4326 (`lat/lon`).
- **Attributes used.** `BEDS` (staffed beds; the HIFLD `-999` "unknown" bed sentinel is masked to missing). Missing beds are uncommon nationally (415 of 7,966 hospitals, 5.2%; total known beds 1,087,608, median 70), but are concentrated enough that 9 zones have over half their beds unknown and are flagged where the bed result depends on imputation. The `TRAUMA` designation is recorded but the trauma-subset analysis is deferred to future work.
- **Vintage.** A single snapshot dated approximately 2024. HIFLD is not a time series; hospitals open and close over time, but the snapshot reflects one cross-section. We therefore treat the hospital layer as static and analyze population time variation against a fixed hospital geography (see Limitations).

Population (ORNL LandScan Global, ambient)

The demand surface is LandScan Global ambient population (ORNL): a gridded, roughly 30 arc-second (about 1 km at the equator) raster of 24-hour average ("ambient") population, distributed annually (one GeoTIFF per year), native EPSG:4326, globally complete, so Alaska, Hawaii, and the territories receive real values. The LandScan record covers 2018 through 2024; the headline analysis uses **2020** to align with the residential surface and the commuting-zone vintage, and the other years support a temporal sensitivity check against the fixed hospital geography.

LandScan estimates where people are over a day, incorporating workplaces, roads, and land cover, rather than where they sleep. This property is what we exploit: it is a complementary demand surface whose timing differs from the residential denominator that accessibility analyses use by default. We do not claim it is the single correct surface for acute care. Many acute events (nocturnal myocardial infarction, stroke, falls among older adults) occur at or near home, and admissions reflect a mix of daytime and residential exposure; ambient population captures the daytime component that residential surfaces miss, and our point is that the choice between the two materially changes the verdict, not that one is uniformly right.

Population (NASA SEDAC GPW v4.11, residential, for contrast)

To isolate the effect of the ambient choice, we replicate the entire analysis with a conventional residential (nighttime) population surface: the Gridded Population of the World, version 4.11 (GPWv4.11) from NASA's Socioeconomic Data and Applications Center (SEDAC; CIESIN, Columbia University) (Doxsey-Whitfield et al., 2015). GPW is provided natively at 30 arc-seconds (about 1 km at the equator), the same grid as LandScan Global, so the two surfaces align essentially cell for cell and the contrast is a clean grid-versus-grid swap with no areal-interpolation step. Both rasters are read identically; only the intensity surface changes.

Two differences, not one: a controlled-contrast caveat. GPW and LandScan differ in the property we wish to study, residential versus ambient timing, but they also differ in construction. GPW is areal and proportional (census counts are spread uniformly within each input administrative unit, so its effective resolution is the input unit size), whereas LandScan is dasymetric and modeled (counts are redistributed within units using roads, land cover, and imagery). A raw LandScan-versus-GPW difference therefore conflates ambient versus residential with modeled versus unmodeled. To separate the two we add a third surface as a control: **WorldPop** (UN-adjusted, ~1 km, 2020), which is dasymetric and modeled like LandScan but residential like GPW. If the ambient effect were a modeling artifact, WorldPop would behave like LandScan; if it is about day-versus-night timing, WorldPop behaves like GPW (Results confirm the latter). WorldPop is distributed per country and we use the US file, so Puerto Rico is not covered by this control.

Analysis windows (USDA commuting zones)

The unit of analysis is the commuting zone (CZ), a cluster of counties tied together by commuting flows (Tolbert & Sizer, 1996). Commuting zones are chosen over metropolitan and micropolitan statistical areas (CBSAs) for three reasons that matter for this test.

1. **National coverage including rural.** Commuting zones partition the entire United States; every county belongs to exactly one zone. Nonmetropolitan rural areas, where hospital access deserts are most severe,

therefore remain in sample. CBSAs, by contrast, exclude all outside-CBSA rural counties.

2. **Movement-defined, matching ambient population.** Commuting zones are delineated from where people travel, the same behavioral basis as LandScan's ambient surface, so the window and the demand surface share a logic.
3. **Exogenous to hospital locations.** Unlike Dartmouth Hospital Service and Referral Regions, which are drawn from hospital-utilization patterns and so are partly endogenous to the thing we test, commuting zone boundaries do not depend on where hospitals are, which avoids circularity.

We use the USDA Economic Research Service 2020 commuting-zone delineation, which groups the 3,222 US and Puerto Rico counties into 598 contiguous labor markets following the Fowler, Rhubart, and Jensen methodology (Fowler et al., 2016; Fowler & Jensen, 2024). CZ geometries are built by dissolving US Census county polygons (TIGER/Line) using the ERS county-to-CZ crosswalk. The 2020 vintage is chosen to match the 2020 population rasters (below). One reconciliation is required: the 2020 crosswalk keys Connecticut (a single commuting zone spanning the whole state) to the state's 2022 planning-region FIPS codes, which are not present in the 2020 county boundary file (Connecticut there is still its eight historical counties); we map Connecticut's counties to that zone so the state is retained, giving the full set of 598 commuting zones.

As a windowing-comparability check we additionally re-run the entire analysis on **Core-Based Statistical Areas** (939 metropolitan and micropolitan CBSAs, Census TIGER 2020 cartographic boundaries). CBSAs are the more familiar unit but cover only urbanized cores and their commuting fringes, omitting the outside-CBSA rural areas; comparing the two window sets tests whether the findings depend on the partition rather than on the data. The CBSA run is reported as a sensitivity, not the headline.

Coordinate reference systems

Population is read in EPSG:4326. Within each commuting zone, the points, the boundary polygon, and the population raster are projected to a local UTM zone (chosen from the zone centroid) so that the distance-based K-function and the area integral of the intensity are computed in meters with minimal distortion.

Derived inputs

File (gitignored, under `data/`)	Produced by	Contents
`hospitals.gpkg`	`acquire/01_hospitals.py`	open hospital points + `beds`, `is_trauma`
`commuting_zones.gpkg`	`acquire/02_commuting_zones.py`	CZ boundary polygons (counties dissolved by USDA crosswalk)
`landscan/landscan-global-<year>.tif`	`acquire/03_landscan.py`	ambient population rasters (LandScan Global)
`gpw/gpw_v4_<year>.tif`	`acquire/04_gpw.py`	residential population rasters (NASA SEDAC GPWv4.11)
`cbsa.gpkg`	`acquire/05_cbsa.py`	939 CBSA windows (Census TIGER 2020), robustness re-run

File (gitignored, under `data/`)	Produced by	Contents
`worldpop/worldpop_2020.tif`	`acquire/06_worldpop.py`	modeled-residential control raster (WorldPop US, ~1 km)

Methods

Setup and notation

Within a single commuting-zone window W (a polygon, projected to local UTM), let $X = \{x_1, \dots, x_n\}$ be the open hospital locations in W. Let $p(u)$ be the population surface (LandScan ambient, or GPW residential) at location $u \in W$, and let b_i be the staffed-bed count of hospital i. We characterize each zone on three orthogonal axes: whether supply is **concentrated** beyond population, whether population is **covered** by nearby hospitals, and whether capacity is **sufficient** per person. Across the three, the population surface is held fixed and the hospitals are the random quantity; the inferential question is always whether the observed hospitals depart from a process tied to population.

Zone inclusion criterion

The inhomogeneous K-function is a second-order (pairwise) summary, so it requires enough hospitals per zone to estimate. Below a small count the Monte-Carlo envelope is degenerate and the test has no power. We therefore compute the concentration tests only for zones with at least **8 open hospitals**. A simulation power analysis (against an inhomogeneous Thomas over-concentration alternative) shows the global test reaches about 67% power at $n = 8$ and 80% at $n \approx 10$, with Type-I near nominal throughout, so we treat 8 as a permissive floor and confirm in Results that the headline is stable when the floor is raised. Of the 598 commuting zones, **262 meet the threshold and 336 do not**, but the excluded zones hold only about **9% of population**, so the concentration analysis still covers roughly **91% of where people are during the day**. The **sufficiency** and **coverage** axes are computed for *every* zone, including sparse ones, since neither requires a minimum hospital count; this is deliberate, because the sparse zones are exactly where access shortfalls are most likely.

Axis 1: Concentration (inhomogeneous K)

The population-proportional intensity

We define the null intensity as population, rescaled so its integral over the window equals the observed hospital count:

$$\lambda(u) \;=\; n\,\frac{p(u)}{\int_W p(v)\,dv}, \qquad u \in W.$$

By construction $\int_W \lambda = n$, so a process drawn from λ produces n hospitals in expectation and the inhomogeneous K below is comparable to its Poisson value πr^2. Because λ is **supplied by the population raster**, there is no kernel bandwidth to choose and no leave-one-out intensity to floor; the instability that arises when λ is estimated from sparse points does not occur. A small positive floor is applied only to populated-by-a- hospital-but-zero-population cells, to keep $1/\lambda$ finite.

The estimator

We use the inhomogeneous K-function (A. J. Baddeley et al., 2000):

$$\widehat{K}_{\text{inhom}}(r) \;=\; \frac{1}{|W|}\sum_i \sum_{j \neq i} \frac{\mathbf{1}\{\|x_i - x_j\| \le r\}\, e(x_i, x_j)}{\lambda(x_i)\,\lambda(x_j)},$$

with $e(\cdot,\cdot)$ the Ripley isotropic edge-correction weight. Under the null that hospitals follow population, $K_{\text{inhom}}(r) \approx \pi r^2$; values above indicate hospitals more clustered than population warrants (over-concentration), values below a more regular-than-population arrangement.

Bed-weighted concentration

To test whether *capacity* (not just buildings) is distributed proportional to population, we weight each hospital by its bed count. Using the per-point intensity identity, a weight w_i with intensity Λ equals an ordinary inhomogeneous K with per-point intensity $\Lambda(x_i)/w_i$, where $\Lambda(u) = (\sum_i w_i)\, p(u) / \int_W p$ is the bed-mass intensity. This follows the mark-weighted K tradition (Giuliani et al., 2014; Penttinen et al., 1992) and requires no custom estimator. Missing bed counts (the HIFLD `-999` sentinel, 5.2% of hospitals nationally) are imputed with the zone median; zones where most beds are unknown are flagged and their bed verdict treated as unreliable.

What is permuted, and the envelope

The population surface is fixed; the hospitals are resampled. Each null realization is a fresh hospital pattern drawn from the population-proportional inhomogeneous Poisson process rpoispp(λ). With `nsim = 999` simulations we form the **global rank envelope test** (extreme-rank-length, ERL) of Myllymäki et al. (2017), which controls the error rate over the whole r-curve and returns one global p-value per zone, rather than the per-distance pointwise envelope (which is miscalibrated: too narrow at small `nsim`, far too conservative at 999). For the bed test the same simulated locations carry bed counts resampled from the zone's own hospitals.

Axis 2: Coverage (population-weighted distance to care)

Concentration measures the arrangement of hospitals relative to each other; **coverage** measures whether population is near care. For each populated ~1 km cell we compute the distance to the nearest hospital and summarize the population-weighted fraction beyond distance d, $C(d)$, across a grid of d. $C(d)$ is compared against the same null (hospitals resampled proportional to population): a curve **above** the envelope means more population is stranded far from care than a population-proportional placement would produce (**under-served**); **below** means better-than-proportional coverage.

Cross-border (buffered) coverage

A person near a zone edge may be served by a hospital in a neighboring zone, so restricting coverage to in-zone hospitals would invent false edge deserts. The coverage axis therefore uses every hospital within the zone **or within a 50 km buffer** of it. The null resamples only the **in-zone** hospitals (the within- zone allocation question) while holding the **out-of-zone buffer hospitals fixed**, so observed and null both credit cross-border access. We report the descriptive share of population beyond 10, 25, and 35 miles of a hospital (the 35-mile figure corresponds to the CMS Critical Access Hospital distance rule); the test itself is threshold-free. Distances are Euclidean, a deliberate simplification discussed in the limitations.

Axis 3: Sufficiency (capacity per capita)

Both tests above condition on the observed hospital (or bed) total, so neither asks whether there is *enough* capacity. We measure that with the standard metric, **staffed beds per 1,000 population**, reported against the US national average (~2.8 (KFF (Kaiser Family Foundation), 2020)) and the OECD average (~4.3 (OECD, 2023)). It is a descriptive ratio, computed for every zone (including the sparse zones the concentration tests skip), under both ambient and residential population. Folding sufficiency into the K statistic would confound it with concentration (over- supply and over-clustering shift the curve the same way), so it is kept as a separate, orthogonal coordinate.

A diagnostic typology

The three axes combine into a per-zone diagnosis that a single accessibility score cannot provide: low sufficiency with under-served coverage indicates a genuine **shortage**; adequate sufficiency with over-concentration and under- served coverage indicates **maldistribution** (enough supply, clumped, fringe stranded); adequate sufficiency with proportional concentration but under-served coverage indicates a **geographic gap** (remote population); high sufficiency with over-concentration and over-served coverage indicates **redundancy**. The national results are reported as the distribution of zones across this typology, which maps each failure mode to a different policy response.

Two-level inference

Within each zone, the global ERL envelope controls error across the r (or d) grid. Across the several hundred zones, running one test per zone is a multiple-comparison problem, so we apply the **combined** global envelope test (Mrkvička et al., 2017; Myllymäki et al., 2017) to control the family-wise error rate across zones, reusing each zone's saved simulated curves rather than re-simulating. This is run separately for each axis.

Local decomposition and maps

For zones flagged on concentration, `localKinhom` decomposes the global K into a per-hospital contribution, mapping which hospitals drive the clustering. For coverage, the per-cell distance field maps where under-served population sits. Together these convert a global verdict into an interpretable geography.

Ambient versus residential

Every quantity is computed twice, once with the LandScan ambient λ and once with the GPW residential λ, holding the hospital data, window, buffer, edge correction, grids, and seeds fixed. The reported contrast is the set of zones whose diagnosis changes between the two demand surfaces.

Validation and robustness

We validate the estimators and probe every discretionary choice; the outcomes are in Results.

Type-I calibration. For each estimator we draw many patterns under the null (hospitals proportional to population, beds resampled as in the engine) and test each against the others, which estimates the false-positive rate. The facility test also reproduces standard spatstat `Kinhom` behavior; the newer bed and coverage estimators are the ones this most needs to confirm.

Power and the inclusion threshold. We estimate the concentration test's power as a function of the hospital count n against a meaningful over-concentration alternative, an inhomogeneous Thomas cluster process whose first-order trend follows population but which adds genuine second-order clustering (a $\texttt{pop}^{\gamma}$ first-order shift is not a valid alternative here, since the intensity-reweighted K is designed to ignore it). For each n we build one null pool of curves and test many alternative draws against it. The $\geq$ 8-hospital floor for the concentration test is set from this curve and reported with a sensitivity re-run at 8, 10, and 15 hospitals.

Buffer-distance sensitivity. Because the coverage null credits hospitals within a fixed buffer, we recompute the coverage test at 25, 50, and 100 km to confirm the desert verdicts are not an artifact of the buffer choice.

Modeled-versus-areal control (WorldPop). LandScan (ambient) is modeled while GPW (residential) is areal, so the ambient-versus-residential contrast could be confounded by construction. We repeat the analysis with WorldPop, a surface that is modeled like LandScan but residential like GPW; whether it patterns with the ambient or the residential surface isolates the two effects.

Windowing comparability (CBSA). We re-run the entire analysis on 939 CBSA windows to test whether the findings depend on the commuting-zone partition.

Implementation and reproducibility

The reference engine is R / `spatstat` (`Kinhom`, `localKinhom`, `rpoispp` on a pixel-image intensity, and the `GET` package for the global test), orchestrated across zones and population surfaces by a Nextflow pipeline (local, SLURM, and AWS Batch profiles). Each run writes envelope curves, verdicts, global p-values, the saved simulations, and metadata; code is in `code/` and is not part of the submitted manuscript.

Results

All results use the 2020 vintage (LandScan ambient and GPWv4.11 residential population, USDA 2020 commuting zones, HIFLD hospitals). Point-process tests use 999 simulations and the global extreme-rank-length (ERL) envelope, corrected for family-wise error across zones. Numbers are generated from `output/summary.csv` and the across-zone tests; per-zone diagnoses are in `output/typology_*.csv`.

Sample

The 598 US commuting zones contain 7,966 open hospitals (1,087,608 known staffed beds). Every zone receives a **sufficiency** value and, where it has at least one hospital, a **coverage** test (586 zones); the 12 zones with no hospital are reported separately. The second-order **concentration** test is computed for the 262 zones with at least 8 hospitals, which together hold about 91% of the population. Each analysis is run twice, under ambient and residential population.

Concentration

Whether hospital locations are more clustered than population depends entirely on the population surface. Under **residential** population, **15 of 262** zones are significantly over-concentrated on facilities (combined family-wise $p = 0.002$) and **13 of 262** on beds ($p = 0.001$). Under **ambient** population, the signal vanishes: **0 of 262** zones on either facilities ($p = 0.475$) or beds. In other words, hospital clustering that looks excessive against where people sleep is fully consistent with where people are during the day. The over-concentrated (residential) set is led by large and mid-size metros (for example Dallas, San Diego, Philadelphia, Orlando, Las Vegas, New Orleans). The effect is substantial where present: expressed through the variance- stabilized L-function ($L = \sqrt{K/\pi}$), the flagged zones cluster at a 10 km scale a median of 2.9 times more tightly than a population-proportional placement (interquartile range roughly 1.5 to 11 times, and higher in a few sparse zones such as Anchorage), so these are large departures, not marginal ones.

The bed-weighted test is not redundant with the facility test: capacity and building count diverge. Of the residential-population zones flagged, 9 are over-concentrated on both facilities and beds, 6 on facilities only (Brownsville, Dallas, Dayton, Erie, Johnson City, San Diego, where many similarly sized hospitals cluster without concentrating capacity), and 4 on beds only (Allentown, Augusta, Jacksonville, Saginaw, where a few large hospitals concentrate capacity even where building locations are not over-clustered). Measuring capacity therefore flags zones a facility-count analysis would miss. The bed weighting imputes the 5.2% of hospitals with an unknown bed count; the 9 zones where more than half of beds are unknown are flagged and excluded from bed claims.

Coverage

The pattern reverses on coverage. Under **ambient** population, **6 of 586** zones have populations significantly farther from care than a population- proportional placement would leave them (combined $p = 0.001$): Amarillo, Dallas, Detroit, Houston, San Diego, and San Juan (PR). Under **residential** population, **0 of 586** survive family-wise correction ($p = 0.126$). The ambient lens reveals access gaps that the residential surface hides. Because coverage credits hospitals within 50 km across zone boundaries, these are not edge artifacts. The magnitudes are meaningful: at the distance of peak deviation, each desert leaves an extra 9 to 37 percentage points of its population beyond reach relative to the null (largest in Amarillo), and Amarillo's population-weighted mean distance to care runs about 3 km above what a population-proportional placement would give.

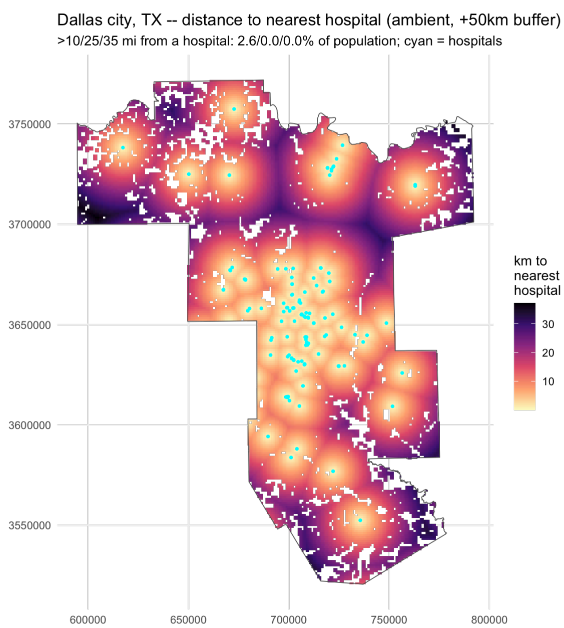

Figure 1: Distance from each populated cell to the nearest hospital in the Dallas commuting zone (ambient population, 50 km cross-border buffer). Warm cells are populated areas far from care; cyan points are hospitals. The under-served population sits in the outer ring that a metropolitan-only window would exclude.

Sufficiency

Zone-level capacity is right-skewed: the median zone has 4.6 (ambient) / 4.4 (residential) staffed beds per 1,000, well above the US population-weighted average of ~2.8 (KFF (Kaiser Family Foundation), 2020), because many low-population rural zones carry high per-capita capacity while dense metros sit low, the documented rural-urban capacity gradient (Hegland et al., 2022). Below the US reference fall **88** zones under ambient and **114** under residential population; below the OECD reference (~4.3), 251 and 289 respectively. The denominator matters: commuter-destination zones look better supplied on residential population and worse on ambient, and the reverse for bedroom communities.

The diagnostic typology

Combining the three corrected axes classifies each zone (Table 1; national maps below).

Diagnosis	Ambient	Residential
no hospital	12	12
shortage	0	0
maldistribution	0	0
geographic gap	6	0
capacity shortfall	76	102
redundancy	0	13
well-matched	503	470

Two results stand out. First, **over-concentration and under-service do not co-occur** in the same zone (maldistribution = 0 under either surface). This is partly structural and should be read as such: concentration is tested only in the 262 hospital-dense zones, whereas the deserts are rural zones that mostly fall below the 8-hospital floor, so the two diagnoses are drawn from largely disjoint sets of zones and their non-overlap is closer to expected than surprising. It is still informative that no dense zone is simultaneously clustered and stranded, but we do not read it as a strong empirical discovery. Second, the dominant flagged category flips with the demand surface: residential population yields a **redundancy** story (13 zones, clustering beyond population) and no coverage gaps, whereas ambient population yields a **geographic-gap** story (6 zones) and no redundancy.

Ambient versus residential

44 of 598 zones change diagnosis between the two surfaces. The clearest cases are metros that are simultaneously over-concentrated against residen-

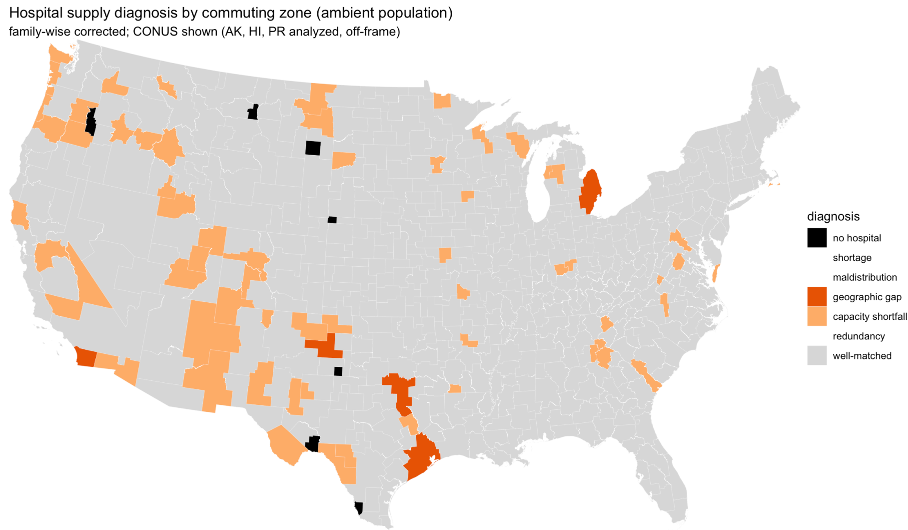

Figure 2: Diagnostic typology of hospital supply by commuting zone under **ambient** population (CONUS; Alaska, Hawaii, and Puerto Rico analyzed but off-frame). Grey is well-matched; orange is capacity shortfall; dark orange is a coverage gap; black is a zone with no hospital.

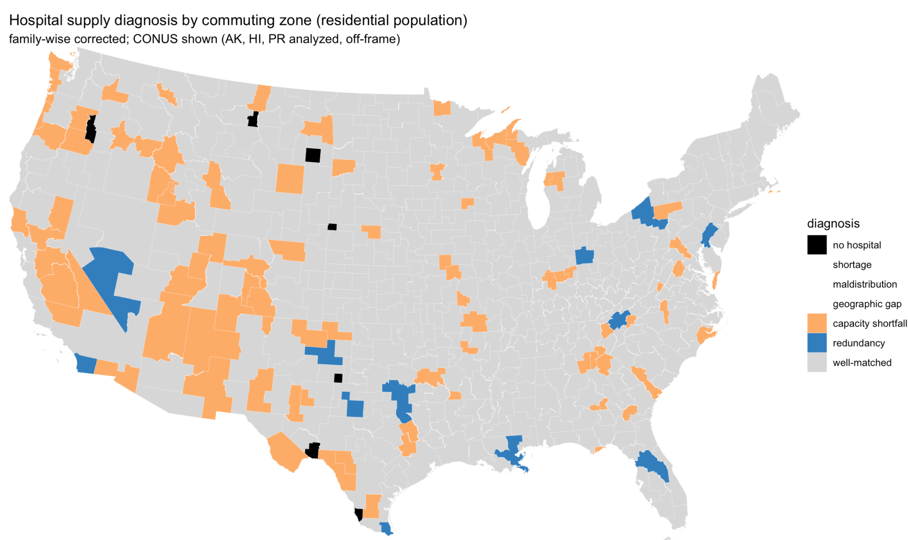

Figure 3: The same diagnosis under **residential** population. Over-concentration (blue, redundancy) appears that is absent under the ambient surface, and the coverage gaps disappear.

tial population and under-served against ambient population, so their diagnosis moves from *redundancy* (residential) to *geographic gap* (ambient): Dallas, San Diego, and Amarillo are in both flagged sets. Dallas is illustrative (panels below): its hospitals cluster beyond nighttime population yet leave daytime population comparatively far from care. The demand surface does not refine the conclusion; it reverses it.

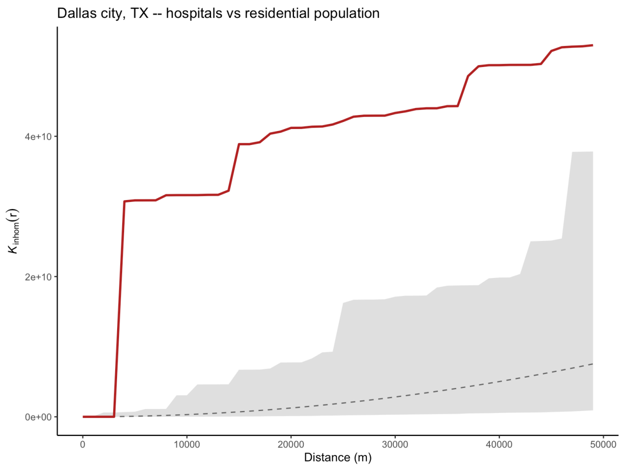

Figure 4: Dallas, residential population: the observed inhomogeneous K (red) rises above the simulation envelope (grey), i.e. hospitals are more clustered than residential population predicts (over-concentrated).

This reversal is driven by the day-versus-night timing of the demand surface, not by how the surface is built. LandScan (ambient) is a dasymetric, modeled product while GPW (residential) is areal, so a modeled-versus-areal difference could in principle confound the ambient-versus-residential contrast. We rule that out with a third surface, WorldPop, which is modeled like LandScan but residential like GPW. Under WorldPop, hospital concentration behaves like the residential (GPW) surface, not the ambient one: 18 of 258 zones are significantly over-concentrated on facilities (combined $p = 0.001$) and 17 on beds, with 0 of 576 zones under-served on coverage ($p = 0.477$). Both residential surfaces, whether areal (GPW) or modeled (WorldPop), yield over-concentration and no coverage gaps, while the ambient surface yields the reverse. The effect therefore tracks whether population is counted by day or by night, not the construction of the raster. (WorldPop is US-only, so Puerto Rico and a small number of coverage-gap zones are absent, leaving 258 rather than 262 concentration zones; this does not affect the comparison.)

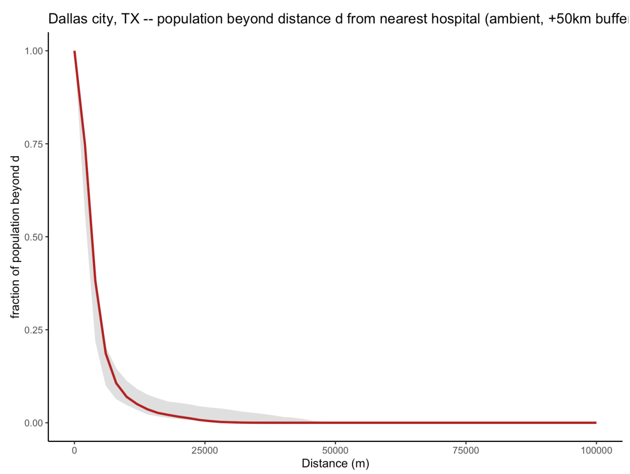

Figure 5: Dallas, ambient population: the observed coverage curve (red), the share of population beyond distance d from a hospital, rises above the envelope, i.e. more daytime population is far from care than a population-proportional placement would leave (under-served).

Validation and robustness

Type-I calibration by simulation (target 0.05) gives facilities 0.040, beds 0.043, and coverage 0.040, confirming the estimators, including the newer bed and coverage constructions, are well behaved. A power analysis against an inhomogeneous Thomas over-concentration alternative (cluster scale 2.5 km) shows power rising from 0.67 at 8 hospitals to 0.80 at 10 and 0.99 at 30, with Type-I near nominal throughout, so the 8-hospital floor is permissive rather than lax. The concentration result is stable to that floor: re-running the residential facility test at minimum 8, 10, and 15 hospitals flags 15, 14, and 9 zones (combined $p = 0.002, 0.002, 0.028$) and the bed test 13, 12, and 9, so raising the threshold only shrinks the eligible pool without overturning the finding.

The coverage axis has its own power analysis. Against a manufactured desert (a zone whose hospitals are drawn proportional to population but only within a central core, stranding a set fraction of the population in the periphery), Type-I is near nominal (about 0.05) throughout, and power depends on the desert's geometry: in a geographically distinct desert such as San Diego's backcountry the test reaches power 0.99 once 20% of the population is stranded, whereas in a monocentric zone such as Amarillo, where a population-proportional placement already leaves the sparse periphery far from care, power is lower (about 0.37 at 40% stranded). The coverage test is therefore conservative, and the six zones it flags family-wise are strong rather than marginal detections. Finally, the concentration verdicts do not depend on using a Poisson (random-count) null: re- running eight representative zones with a fixed-count binomial null (exactly

n hospitals placed proportional to population) reproduces every verdict (8 of 8 agree), with p-values differing by at most 0.16 and only in zones far from the 0.05 boundary.

The coverage (desert) results are invariant to the cross-border buffer distance. Recomputing the coverage test for the ambient under-served zones at 25, 50, and 100 km buffers leaves every zone's verdict unchanged: Amarillo, Dallas, Detroit, Houston, and San Juan remain significantly under-served at all three distances, and San Diego remains borderline throughout. Crediting hospitals as far as 100 km across zone boundaries therefore does not dissolve the deserts, so they are not an artifact of the 50 km default.

Nor are the deserts an artifact of straight-line distance. For all six under-served zones we routed a population-weighted sample of each zone's most distant populated cells to the nearest hospital on the OpenStreetMap driving network (OSRM) and compared the road distance to the Euclidean distance. Road distance exceeds Euclidean distance in every zone, by a median factor of 1.13 (Amarillo) to 2.62 (Houston), so travel distance makes the deserts worse, not better: the stranded populations remain at least 10 km from care by road, and in the two genuinely rural deserts (Amarillo and San Diego's backcountry) the most distant decile sits 30 to 70 km away by road. Because road detours only lengthen these distances, Euclidean distance is a conservative proxy here and the deserts are not a straight-line artifact.

Re-running the entire analysis on 939 CBSA windows shows the concentration result is robust to the partition while clarifying the coverage result. Under CBSAs, concentration behaves exactly as under commuting zones: 0 of 185 hospital-dense CBSAs are over-concentrated against ambient population and 15 of 185 against residential (combined $p = 0.001$). Coverage, by contrast, flags no zones under either surface on CBSAs (0 of 927). This is expected and supportive rather than contradictory: CBSAs cover only urbanized cores and their immediate commuting fringe, so the under-served populations, which sit in the rural areas that commuting zones include but CBSAs omit, lie outside the CBSA windows altogether. The coverage deserts are therefore a genuine property of rural populations that only the whole-country commuting-zone partition captures, not an artifact of the commuting-zone boundaries; and the concentration finding does not depend on the windowing choice at all.

A final check addresses whether the ambient result is mechanically induced by the hospitals themselves. LandScan assigns daytime population to workplaces, including hospitals (staff, outpatients, and visitors), so an ambient cell that contains a hospital is inflated partly because the hospital is there. If that self-induced population were large it could artificially satisfy the population-proportional null and explain why over-concentration disappears under ambient population. It does not. The share of each concentration zone's ambient population falling in the roughly 1 km cells that contain a hospital is small (median 2.7%, at most 8.8%, and 3.2% pooled across zones), and this is an upper bound because those cells also hold ordinary residents and workers unrelated to the hospital. To test the concern directly, we re-ran the concentration axis on a modified ambient surface in which every hospital-containing cell is replaced by its local neighborhood background (the mean of the surrounding cells), stripping the hospital's self-induced population while leaving the surrounding demand field intact. We replace rather than delete the cell because the inhomogeneous K-function weights each hospital by the inverse of the null intensity at its own location, so zeroing that cell would divide by an intensity of almost zero and spuriously inflate clustering. Under this hospital-neutralized ambient surface the result is unchanged: 0 of 262 zones are over-concentrated (combined $p = 0.446$), the same verdict and essentially the same p-value as the unmodified ambient surface (0 of 262, $p = 0.475$). The ambient-versus-residential reversal therefore does not arise from population that hospitals induce at their own locations.

A related question is whether the reversal is merely a restatement of "LandScan is uniformly more urban-core-weighted than GPW," which would make it a mechanical consequence of raster geometry rather than an informative result. It is not. If one surface were the other scaled by a single urban multiplier, then after normalizing each surface to integrate to one over a zone their ratio would be identically one everywhere, because the normalization cancels any global factor. Instead, the normalized ambient-to-residential ratio at hospital locations has a median of 1.73 across the 262 zones (above one in 251 of them) against a residential-weighted baseline of exactly one, and it is higher still (1.94) in the fifteen zones that flip. Hospitals therefore sit disproportionately in cells that carry more of the zone's ambient (daytime) population than its residential population, a structured coincidence between hospital sites and daytime destinations that a uniform reweighting cannot produce. The reversal reflects where hospitals actually are relative to where people are by day, not raster construction.

One asymmetry in the typology deserves emphasis. Concentration and coverage are tested against a null; sufficiency is compared to a fixed bed-per-1,000 threshold and is therefore not a test but a threshold crossing. Its largest typology cell, capacity shortfall, is correspondingly sensitive to that threshold: the count of residential capacity-shortfall zones moves from 38 to 102 to 277 as the cutoff rises from 0.7 times the US average to the US average (2.8 per 1,000) to the OECD average (4.3 per 1,000), whereas the tested categories (redundancy, geographic gap) are unaffected. Sufficiency conclusions should be read as descriptive relative to an explicit, and movable, reference rather than as a statistical verdict, and the beds-per-1,000 denominator inherits the same commuter-destination-versus-bedroom-community ambiguity as the rest of the study.

Conclusion

Summary

We characterized US hospital supply on three orthogonal axes, concentration, coverage, and sufficiency, in every commuting zone, each against a population-proportional null and contrasted between ambient and residential population. The central result is that the demand surface reverses the conclusion: against residential population the country shows significant hospital over-concentration and no coverage gaps, while against ambient population the over-concentration vanishes and genuine coverage gaps appear instead. Forty-four zones change diagnosis between the two surfaces, and over-concentration and under-service are not found together in the same zone (partly structural, since concentration is tested only in the hospital-dense zones). The pattern holds up under scrutiny: it is stable to the hospital-count threshold and the coverage buffer, a WorldPop control shows the reversal is about day-versus-night timing rather than raster construction, neutralizing the population LandScan places at the hospital sites themselves leaves the ambient result unchanged, and a re-run on CBSA windows reproduces the concentration result while confirming that the coverage deserts live in the rural areas only the commuting-zone partition captures.

Limitations

- **Static hospital snapshot, and a vintage gap.** HIFLD is a single cross-section current to about mid-2024 (its records carry 2024 source dates), which we test against 2020 population, commuting zones, and rasters, a roughly four-year gap. The dataset carries no per-facility opening or closing date and no 2020-vintage layer was available to us, so we cannot re-run on a contemporaneous hospital geography. Two considerations bound the concern. The coverage deserts are conservative with respect to this gap: a hospital that opened between 2020 and 2024 and appears in our layer makes a zone look less deserted than it was in 2020, while a closure we cannot see would make it worse, so the six deserts are lower bounds. The over-concentration result is a family-wise finding across 262 zones with large effect sizes (a median tenfold- scale clustering ratio near three), which the low-single-digit-percent net change in the US hospital stock over four years cannot manufacture. A dated HIFLD or CMS panel would still enable a stronger, truly longitudinal design.
- **Ambient is not acute demand.** Ambient population is a better proxy than residential for where acute events occur, but it is still a proxy. True demand also depends on age structure, morbidity, and case mix, which a population surface does not capture.
- **Modeled versus areal construction.** The LandScan (ambient, modeled) versus GPW (residential, areal) contrast could in principle mix tim-

ing with raster construction. A WorldPop control (modeled but residential) resolves this: it behaves like the residential surface, so the effect tracks day-versus-night timing, not construction. A residual limitation is that WorldPop is US-only, so Puerto Rico is absent from that check.

- **Window dependence.** Results are conditional on the commuting-zone partition. The CBSA re-run shows the concentration finding is insensitive to the partition, but the coverage deserts are visible only with commuting zones, because CBSAs exclude the rural fringe where the under-served populations sit. The coverage conclusion is thus specific to a whole-country partition; a metropolitan-only windowing would miss it by construction.
- **Distance, not travel time.** All distances are Euclidean, whereas real access depends on roads, terrain, and traffic. The method answers a question about spatial distribution that complements, rather than replaces, travel-time accessibility models; the reported distance thresholds (10, 25, 35 miles) are straight-line approximations.
- **Containment.** The coverage axis credits hospitals within a 50 km buffer across zone boundaries, which removes false edge deserts, but concentration and sufficiency remain in-zone quantities, so a zone whose residents routinely use a neighbor's hospitals may still be mischaracterized on those two axes. The USDA commuting-zone design (built for self-containment) limits this, and the `CZContainment` measure flags the most exposed zones.
- **Beds missing for some hospitals.** 5.2% of hospitals lack a bed count; sufficiency is therefore a lower bound, and bed-concentration verdicts in the 9 zones where most beds are unknown are imputation-driven and flagged.

Implications

The choice of demand surface is not a technical detail; it reframes the policy question. A residential denominator invites a reading of urban supply as excessive, while an ambient denominator recasts the same metros as places with daytime access gaps. We stop short of prescribing that capacity be removed: statistical over-concentration relative to a population-proportional null is not by itself evidence of inefficiency or inequity, since hospital location also reflects agglomeration, tertiary-referral role, historical siting, and certificate-of-need regulation, none of which this test observes. What the analysis does license is narrower and, we think, more useful: that an accessibility assessment can flip between "over-supplied" and "under-serving" for the same place purely on the demand surface, so the ambient framing deserves weight alongside the residential default in capacity and siting deliberations. The diagnostic typology further separates the distinct failures, too little capacity, capacity in the wrong place, and populations left far from care, each of which calls for a different response.

Reproducibility and data availability

All inputs are public (HIFLD via DataLumos; LandScan Global via ORNL; GPWv4.11 via NASA SEDAC; commuting zones via USDA ERS and Census TIGER). Code and the exact acquisition steps are in the project repository under `code/`.

References

Baddeley, A. J., Møller, J., & Waagepetersen, R. (2000). Non- and semi-parametric estimation of interaction in inhomogeneous point patterns. *Statistica Neerlandica*, *54*(3), 329–350. https://doi.org/10.1111/1467-9574.00144

Baddeley, A., Rubak, E., & Turner, R. (2015). *Spatial point patterns: Methodology and applications with r*. Chapman; Hall/CRC. https://doi.org/10.1201/b19708

Bhaduri, B., Bright, E., Coleman, P., & Urban, M. L. (2007). LandScan USA: A high-resolution geospatial and temporal modeling approach for population distribution and dynamics. *GeoJournal*, *69*(1–2), 103–117. https://doi.org/10.1007/s10708-007-9105-9

Branas, C. C., MacKenzie, E. J., Williams, J. C., Schwab, C. W., Teter, H. M., Flanigan, M. C., Blatt, A. J., & ReVelle, C. S. (2005). Access to trauma centers in the united states. *JAMA*, *293*(21), 2626–2633. https://doi.org/10.1001/jama.293.21.2626

Carr, B. G., Bowman, A. J., Wolff, C. S., Mullen, M. T., Holena, D. N., Branas, C. C., & Wiebe, D. J. (2017). Disparities in access to trauma care in the united states: A population-based analysis. *Injury*, *48*(2), 332–338. https://doi.org/10.1016/j.injury.2017.01.008

Chea, H., Kim, H., Shaw, S.-L., & Chun, Y. (2022). Assessing trauma center accessibility for healthcare equity using an anti-covering approach. *International Journal of Environmental Research and Public Health*, *19*(3), 1459. https://doi.org/10.3390/ijerph19031459

Diggle, P. J., Gómez-Rubio, V., Brown, P. E., Chetwynd, A. G., & Gooding, S. (2007). Second-order analysis of inhomogeneous spatial point processes using case-control data. *Biometrics*, *63*(2), 550–557. https://doi.org/10.1111/j.1541-0420.2006.00683.x

Dobson, J. E., Bright, E. A., Coleman, P. R., Durfee, R. C., & Worley, B. A. (2000). LandScan: A global population database for estimating populations at risk. In *Remotely sensed cities*. https://doi.org/10.1201/9781482264678-24

Doxsey-Whitfield, E., MacManus, K., Adamo, S. B., Pistolesi, L., Squires, J., Borkovska, O., & Baptista, S. R. (2015). Taking advantage of the improved availability of census data: A first look at the gridded population of the world, version 4. *Papers in Applied Geography*, *1*(3), 226–234. https://doi.

org/10.1080/23754931.2015.1014272

Duranton, G., & Overman, H. G. (2005). Testing for localization using micro-geographic data. *The Review of Economic Studies*, *72*(4), 1077–1106. https://doi.org/10.1111/0034-6527.00362

Fowler, C. S., & Jensen, L. (2024). New commuting zone delineation for the u.s. Based on 2020 data. *Scientific Data*, *11*. https://doi.org/10.1038/s41597-024-03829-5

Fowler, C. S., Rhubart, D. C., & Jensen, L. (2016). Reassessing and revising commuting zones for 2010: History, assessment, and updates for U.S. "Labor-sheds" 1990–2010. *Population Research and Policy Review*, *35*(2), 263–286. https://doi.org/10.1007/s11113-016-9386-0

Giuliani, D., Arbia, G., & Espa, G. (2014). Weighting ripley's k-function to account for the firm dimension in the analysis of spatial concentration. *International Regional Science Review*, *37*(3), 251–272. https://doi.org/10.1177/0160017612461357

Gligorić, K., Kamath, C., Weiss, D. J., Bavadekar, S., Liu, Y., Shekel, T., Schulman, K., & Gabrilovich, E. (2023). Revealed versus potential spatial accessibility of healthcare and changing patterns during the COVID-19 pandemic. *Communications Medicine*, *3*, 157. https://doi.org/10.1038/s43856-023-00384-9

Hegland, T. A., Owens, P. L., & Selden, T. M. (2022). New evidence on geographic disparities in united states hospital capacity. *Health Services Research*, *57*(5), 1006–1019. https://doi.org/10.1111/1475-6773.14010

Hsia, R. Y., & Shen, Y.-C. (2011). Rising closures of hospital trauma centers disproportionately burden vulnerable populations. *Health Affairs*, *30*(10), 1912–1920. https://doi.org/10.1377/hlthaff.2011.0510

KFF (Kaiser Family Foundation). (2020). *The U.S. Has fewer physicians and hospital beds per capita than italy and other countries overwhelmed by COVID-19*. KFF. https://www.kff.org/health-costs/the-u-s-has-fewer-physicians-and-hospital-beds-per-capita-than-italy-and-other-countries-overwhelmed-by-covid-19/

Li, B., Long, J., & Li, Y. (2025). Geospatial analysis of healthcare and older adult care institutions in Wuhan: A multimethod approach to assessing spatial equity. *Frontiers in Public Health*, *13*, 1580630. https://doi.org/10.3389/fpubh.2025.1580630

Luo, W., & Qi, Y. (2009). An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. *Health and Place*, *15*(4), 1100–1107. https://doi.org/10.1016/j.healthplace.2009.06.002

Luo, W., & Wang, F. (2003). Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design*, *30*(6), 865–884. https://doi.org/10.1068/b29120

Marcon, E., & Puech, F. (2003). Evaluating the geographic concentration of industries using distance-based methods. *Journal of Economic Geography*, *3*(4), 409–428. https://doi.org/10.1093/jeg/lbg016

Marcon, E., & Puech, F. (2010). Measures of the geographic concentration of industries: Improving distance-based methods. *Journal of Economic Geography*, *10*(5), 745–762. https://doi.org/10.1093/jeg/lbp056

Mrkvička, T., Myllymäki, M., & Hahn, U. (2017). Multiple monte carlo testing, with applications in spatial point processes. *Statistics and Computing*, *27*(5), 1239–1255. https://doi.org/10.1007/s11222-016-9683-9

Myllymäki, M., & Mrkvička, T. (2024). GET: Global envelopes in R. *Journal of Statistical Software*, *111*(3), 1–40. https://doi.org/10.18637/jss.v111.i03

Myllymäki, M., Mrkvička, T., Grabarnik, P., Seijo, H., & Hahn, U. (2017). Global envelope tests for spatial processes. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *79*(2), 381–404. https://doi.org/10.1111/rssb.12172

Ni, J., Qian, T., Xi, C., Rui, Y., & Wang, J. (2016). Spatial distribution characteristics of healthcare facilities in Nanjing: Network point pattern analysis and correlation analysis. *International Journal of Environmental Research and Public Health*, *13*(8), 833. https://doi.org/10.3390/ijerph13080833

OECD. (2023). *Health at a glance 2023: OECD indicators*. OECD Publishing. https://doi.org/10.1787/7a7afb35-en

Penttinen, A., Stoyan, D., & Henttonen, H. M. (1992). Marked point processes in forest statistics. *Forest Science*, *38*(4), 806–824. https://doi.org/10.1093/forestscience/38.4.806

Sims, K., Reith, A., Bright, E., McKee, J., & Rose, A. (2025). LandScan HD: A high-resolution gridded ambient population methodology for the world. *Population and Environment*. https://doi.org/10.1007/s11111-025-00514-6

Stacherl, B., & Sauzet, O. (2023). Gravity models for potential spatial healthcare access measurement: A systematic methodological review. *International Journal of Health Geographics*, *22*, 32. https://doi.org/10.1186/s12942-023-00358-z

Tolbert, C. M., & Sizer, M. (1996). *U.S. Commuting zones and labor market areas: A 1990 update* (Staff Paper AGES-9614). U.S. Department of Agriculture, Economic Research Service. https://ageconsearch.umn.edu/record/278812/

U.S. Department of Homeland Security. (2024). *Homeland infrastructure foundation-level data (HIFLD): hospitals*. https://catalog.data.gov/dataset/hospitals

Wang, F. (2012). Measurement, optimization, and impact of health care accessibility: A methodological review. *Annals of the Association of American Geographers*, *102*(5), 1104–1112. https://doi.org/10.1080/00045608.2012.657146